

Vigneshwaran Siva Sankaran

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EDUCATION

University of California, San Diego

M.S. in Business Analytics

Focus: Machine Learning, Statistical Modeling, and Data-Driven Decision Systems

La Jolla, CA

Jul 2025 – Dec 2026

Indian Institute of Technology, Madras

Diploma in Data Science and Programming

Chennai, India

Aug 2020 – Dec 2022

TECHNICAL SKILLS

Programming: Python, SQL, Java, C/C++, JavaScript

Machine Learning: CNN, RNN, LSTM, RAG, Sequence Modeling, Feature Engineering, Model Training, Inference Pipelines, Hyperparameter Tuning, Error Analysis

Data Science & Statistics: Regression Models, Time Series Forecasting, Probability, Statistical Inference, Exploratory Data Analysis

ML Frameworks: TensorFlow, scikit-learn, PyTorch

Data Processing & Analytics: Pandas, Polars, NumPy, KPI Design, Analytical Pipelines, Spark (PySpark)

Systems & Infrastructure: AWS, Docker, K2, Git, ETL Pipelines, Batch and Real-Time Workflows

EXPERIENCE

RSKD Labs Pvt Ltd.

Innovation & Development Engineer

Chennai, India

Jul 2023 – Jun 2025

- Engineered and productionized a **multi-module AI system** for **security-critical autonomous applications**, developing **end-to-end ML pipelines** spanning **data ingestion**, **model inference**, **RAG-based retrieval**, and **LLM-driven reasoning** across **10+ system components**.
- Built real-time, frame-by-frame **computer vision inference pipelines** using **CNN-based perception models**, achieving **600 ms latency at 30 FPS** while optimizing **model accuracy**, **inference throughput**, and **runtime stability**.
- Designed and owned **ML infrastructure** and **MLOps workflows** on **AWS**, enabling **continuous model deployment**, **automated retraining**, and **real-time / batch inference** via **CI/CD** and **Dockerized services**.
- Optimized **ML inference and data-processing pipelines**, reducing **end-to-end processing latency** by **30%** and increasing **production value** by **25%** through **model-level and pipeline-level optimizations**, processing **600k - 800k records**

Zoho Corporation

Member Technical Staff Intern

Kattankulathur, India

Jan 2023 – May 2023

- Worked on **production-grade backend APIs** for Zoho Payroll, supporting **1M+ users** and **processing tens of millions of payroll records** in a high-availability financial system.
- Optimized SQL queries** and backend data-access paths, **reducing query execution time** by **15%** and improving **load latency** from **5s to 2s** in production workflows.
- Implemented **scalable backend data-processing logic** within payroll pipelines, ensuring **low-latency**, **correctness**, and **reliability** under high-volume transactional workloads.

SELECTED PROJECTS

AI-IaSS — AI-Driven Infrastructure Cost & Security Optimization (Hackathon Winner)

Jan 2026

Python, SQL, LLM Agents, AWS

- Built an **LLM-driven decision-support system** translating **natural-language infrastructure requirements** into **cloud architecture recommendations**, evaluating **100+ candidate configurations** per request.
- Designed **agent-based cost modeling and optimization pipelines**, simulating trade-offs across **compute**, **storage**, and **networking resources**, achieving **20–30% estimated cost reduction** over baseline configurations.
- Integrated **automated security risk analysis**, analyzing **50+ security rules** to surface **cost–security trade-offs**.

RiboSense — Genomic Sequence Classification

Jul 2025

Python, TensorFlow, BiLSTM, Pandas

- Built a **deep learning ML model** for large-scale **sequence detection and classification**, trained on **1M+ sequences** aggregated across **10+ datasets**.
- Designed a **tile-based sequence modeling pipeline** using **250bp windows**, combining **k-mer representations** and learned **embeddings** to model **100K+ token-length variable sequences** with **BiLSTM architectures**.
- Achieved **0.98 AUC-ROC** through systematic **data preprocessing**, **deduplication**, **hyperparameter tuning**, and **error analysis**, validating generalization on benchmark and real-world test cases.